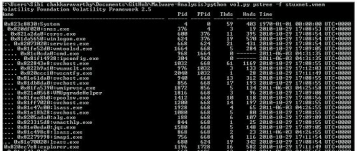

Figure 5: Screenshot of the *pstree* plugin

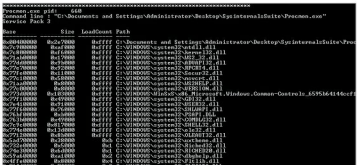

Figure 6: Screenshot of the *dlllist* plugin which scans for hidden processes

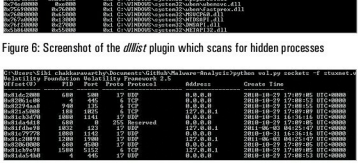

Figure 7: Screenshot of the *sockets* plugin which shows the active TCP connections

3. *Psscan*: This plugin scans the hidden/inactive/unlinked processes and is used for malware analysis. The syntax is:

```
python vol.py psscan -f Stuxnet.vmem
```

4. *Pstree:* This plugin displays the running process in tree form. The syntax is:

```
python vol.py pstree -f Stuxnet.vmem
```

5. *DLL list:* This displays the DLLs used by all the processes. Here, Figure 6 denotes the screenshot of the process ID: 660 and its associated DLLs. The syntax is:

```
python vol.py dlllist -f Stuxnet.vmem
```

6. *Sockets*: This plugin is used to find out the listening socket connections during the time of the memory dump. The syntax is:

```
python vol.py sockets -f Stuxnet.vmem
```

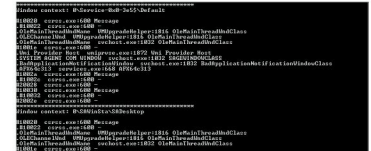

Figure 8: Screenshot of the *wintree* plugin

7. *Timeliner*: This creates a timeline from various artefacts in memory. The syntax is:

```
python vol.py timeliner -f Stuxnet.vmem
```

8. *Wintree*: This prints the Z-Order Desktop Windows tree. The syntax is:

```
python vol.py wintree -f Stuxnet.vmem
```

Present day malware are stealthier and remain hidden during dynamic behaviour analysis. In order to detect such types of malware and their behaviour, run time memory inspection can be carried out. Malware, including rootkit, traverses the RAM; hence the Volatility framework helps us to inspect the live memory of any operating system. This can help us to possibly detect some advanced malware, which is very persistent in its behaviour. END

References

[1] *http://www.volatilityfoundation.org/#!24/c12wa*
[2] *https://github.com/sibichakkaravarthy/Malware-Analysis*
[3] *https://github.com/volatilityfoundation/volatility/wiki/Memory-Samples*
[4] *https://github.com/volatilityfoundation/volatility*
[5] *http://resources.infosecinstitute.com/memory-forensics-and-analysis-using-volatility/*

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